JOTHISHRI B

Full Stack Developer | React, TypeScript, Python

Taramani, Chennai - 600113 | 9360216541 | jothishri03@gmail.com | github.com/Jothi-shri | linkedin.com/in/jothishrib3

PROFESSIONAL SUMMARY

Full Stack Developer with 1 year of hands-on experience building React and TypeScript applications, including real-time monitoring UIs for robotics platforms, backed by FastAPI/Flask REST APIs and PostgreSQL/MongoDB. Skilled in component-driven architecture, React Hooks, state management, and REST/MQTT data integration for live dashboards, with a focus on responsive, high-performance interfaces for data-dense, real-time products.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Responsive Web Design, Component-Driven Development, Design Systems

React: Reusable Components, Hooks, State Management (Zustand), React Router, Performance Optimization

Backend & Databases: Python, FastAPI, Flask, RESTful APIs, JWT Authentication, SQL, PostgreSQL, MySQL, MongoDB, SQLAlchemy, Alembic

Visualization: Leaflet, Three.js, Konva

API Integration: RESTful APIs, MQTT (real-time telemetry), WebSocket / real-time data handling

Testing: Jest (unit testing), Manual Testing

Tooling: Git, GitHub, GitHub Actions, CI/CD, Docker, Vite, npm, Postman, Vercel, Netlify, Render

WORK EXPERIENCE

Web Developer | Navic - Startup Incubated at IITM *Nov 2025 - Present*

- Built React (TypeScript) frontend applications for Autonomous Mobile Robots (AMRs), including Greeting and Shopping Assistance Robots, delivering real-time monitoring and control dashboards used for live fleet operations.
- Designed and maintained a reusable component library for robot status, mission execution, and navigation UI, cutting duplicate UI code across dashboards and speeding up new feature delivery.
- Rendered live SLAM maps, LiDAR scans, battery status, and obstacle detection on responsive, cross-browser interfaces for real-time robot visualization and navigation.
- Integrated frontend interfaces with backend services over MQTT/REST for continuous robot telemetry, applying systematic debugging and performance optimization to keep dashboards responsive under real-time data load.
- Wrote unit tests with Jest and performed manual testing of dashboard features to catch issues before release.
- Collaborated through GitHub pull requests and version-controlled workflows to ship features and fixes reliably.

PROJECTS

Fleet Management - Multi-Robot Operations Dashboard *Team Project*

- Built a responsive React + Leaflet + Tailwind CSS frontend for real-time multi-robot fleet monitoring, navigation, and action management, backed by RESTful APIs in FastAPI.
- Designed reusable, interactive dashboard components for fleet tracking and robot status; integrated MQTT for live robot communication and tuned UI rendering for consistent performance under continuous data updates.
- Managed PostgreSQL and MongoDB with SQLAlchemy; implemented Alembic migrations and data seeding.

Autonomous Mobile Robot (AMR) - Navigation System *Team Project*

- Built React + Leaflet + Konva interfaces for real-time AMR navigation, mission planning, and robot visualization, with consistent component behavior across views.
- Designed the SQL schema and Flask APIs on the backend to support robot communication and autonomous mission execution end-to-end.

SHOBOTFLOW - No-Code/Low-Code Deployment Platform *Team Project*

- Built a no-code/low-code UI for Docker image and container registry management, simplifying image build, tagging, and deployment workflows for developers.
- Designed reusable deployment workflows following CI/CD-style build, tag, deploy stages; version-controlled with Git/GitHub and deployed on Vercel.

SaaS Workspace Platform - Full-Stack Web Application | saasproject-app.netlify.app *Personal Project*

- Built a full-stack workspace app for projects, tasks, team members, analytics, notifications, and settings using React 19, Vite, Tailwind CSS v4, Zustand, and React Router.
- Developed a FastAPI backend with SQLAlchemy 2, Alembic, and PostgreSQL exposing REST endpoints for six modules, secured with JWT authentication (HttpOnly refresh cookie) and protected routes.
- Deployed the frontend on Netlify and the API on Render, configuring CORS, cross-site cookies, and environment variables.

EDUCATION

Bachelor of Engineering, Computer Science | Mahendra Institutions (affiliated to Anna University) *Apr 2025*

GPA: 7.5 | **Languages:** Tamil, English